

transfer of proprietary company code from within their physical premises, which would not be possible with a cloud based IDE.

Deployment Automation Tools

Deployment automation tools like Jenkins or Ansible help in complete automation of the process of deployment of the application once it has been successfully built. It involves moving the release artefact(s) across environments along with configuration changes, database schema changes, firewalls, proxies and all documentation.

Using deployment tools greatly reduce our efforts during application releases but care should be taken to ensure strict adherence to standards and guidelines without which a lot of issues can crop up in the final stage of the software development process.

Some of the best practices of using deployment automation tools are:

Configuration management

We should aim to use a Configuration management database along with our deployment tool to ensure that the configuration data is correctly persisted, audited and versioned every time there is a change in them. This will help in quickly reverting to a 'happy form' state when something goes wrong in our deployment and we want to revert the application to a previous working state.

Use specific infrastructure

While it is a good practice to always deploy code in production like environment to test against realistic cases, sometimes special use cases like regression tests require infrastructure with higher capabilities. The deployment tool must be configured and profiled in a way that the deployment is done on specific machines based on the use cases and these builds are monitored for performance. 